Revision Notes

CBSE Class 12 Physics • NCERT Part I, Chapters 1–8

Electrostatics • Current Electricity • Magnetism • EMI • AC • EM Waves

How to use these notes. Keep the separate **Formula Sheet** next to you. Each chapter has:

- **A Derivations**: short step outlines of the derivations the board asks. Write them out once from these steps.
- **B Diagrams**: the figures to practise, with the labels that carry marks.
- **C Numerical patterns**: every NCERT exercise grouped into patterns, each with one solved question from the book. The small line under the boxes lists which questions belong to which pattern.
- **D Conceptual questions**: "why" questions with one-line answers. **E Exam tips**. **F JEE extras** (purple; not needed for the board exam).
- **G Pattern spotter**: a one-look table for final revision.

Question numbers match the NCERT textbook (Reprint 2026–27).

Contents

Chapter 1 Electric Charges and Fields **4**

1A Derivations (board favourites) . . . 4
1B Diagrams to practise . . . 5
1C Exercise-wise numerical patterns . . . 5
1D Conceptual questions . . . 6
1E Exam tips . . . 7
1F JEE extras . . . 7
1G Pattern spotter . . . 7

Chapter 2 Electrostatic Potential and Capacitance **9**

2A Derivations (board favourites) . . . 9
2B Diagrams to practise . . . 10
2C Exercise-wise numerical patterns . . . 10
2D Conceptual questions . . . 11
2E Exam tips . . . 11
2F JEE extras . . . 12
2G Pattern spotter . . . 12

Chapter 3 Current Electricity **13**

3A Derivations (board favourites) . . . 13
3B Diagrams to practise . . . 14
3C Exercise-wise numerical patterns . . . 14
3D Conceptual questions . . . 15
3E Exam tips . . . 15

3F JEE extras . . . 15
3G Pattern spotter . . . 16

Chapter 4 Moving Charges and Magnetism 17
4A Derivations (board favourites) . . . 17
4B Diagrams to practise . . . 18
4C Exercise-wise numerical patterns . . . 19
4D Conceptual questions . . . 19
4E Exam tips . . . 20
4F JEE extras . . . 20
4G Pattern spotter . . . 20

Chapter 5 Magnetism and Matter 22
5A Derivations (board favourites) . . . 22
5B Diagrams to practise . . . 22
5C Exercise-wise numerical patterns . . . 23
5D Conceptual questions . . . 23
5E Exam tips . . . 24
5F JEE extras . . . 24
5G Pattern spotter . . . 24

Chapter 6 Electromagnetic Induction 25
6A Derivations (board favourites) . . . 25
6B Diagrams to practise . . . 26
6C Exercise-wise numerical patterns . . . 26
6D Conceptual questions . . . 27
6E Exam tips . . . 27
6F JEE extras . . . 27
6G Pattern spotter . . . 28

Chapter 7 Alternating Current 29
7A Derivations (board favourites) . . . 29
7B Diagrams to practise . . . 30
7C Exercise-wise numerical patterns . . . 30
7D Conceptual questions . . . 31
7E Exam tips . . . 31
7F JEE extras . . . 31
7G Pattern spotter . . . 32

Chapter 8 Electromagnetic Waves 33
8A Derivations (board favourites) . . . 33
8B Diagrams to practise . . . 33
8C Exercise-wise numerical patterns . . . 34

8D Conceptual questions and the spectrum 34
8E Exam tips 35
8F JEE extras 35
8G Pattern spotter 35

Chapter 1 Electric Charges and Fields

1A Derivations (board favourites)

1. Field on the axial line of a dipole

Charges $-q$ at A and $+q$ at B, separation $2a$; point P at distance r from the centre on the axis.
$E_{+q} = \dfrac{q}{4\pi\varepsilon_0(r-a)^2}$ (away from B), $E_{-q} = \dfrac{q}{4\pi\varepsilon_0(r+a)^2}$ (towards A).

Net: $E = \dfrac{q}{4\pi\varepsilon_0}\left[\dfrac{1}{(r-a)^2} - \dfrac{1}{(r+a)^2}\right] = \dfrac{q}{4\pi\varepsilon_0}\cdot\dfrac{4ar}{(r^2-a^2)^2}$.

For $r \gg a$: $\boxed{E = \dfrac{2p}{4\pi\varepsilon_0 r^3}}$ along $\vec{p}$ $(p = 2aq)$.

2. Field on the equatorial line of a dipole

Both charges are at distance $\sqrt{r^2+a^2}$ from P, each giving $E_1 = \dfrac{q}{4\pi\varepsilon_0(r^2+a^2)}$.

Components perpendicular to the axis cancel; components parallel to the axis add: $E = 2E_1\cos\theta$, $\cos\theta = \dfrac{a}{\sqrt{r^2+a^2}}$.

$E = \dfrac{2qa}{4\pi\varepsilon_0(r^2+a^2)^{3/2}}$. For $r \gg a$: $\boxed{E = \dfrac{p}{4\pi\varepsilon_0 r^3}}$, opposite to $\vec{p}$.

3. Torque on a dipole in a uniform field

Forces $+q\vec{E}$ and $-q\vec{E}$: net force zero. They form a couple with arm $2a\sin\theta$.

$\tau = qE\cdot 2a\sin\theta = pE\sin\theta$, i.e. $\boxed{\vec{\tau} = \vec{p}\times\vec{E}}$. (In a non-uniform field there is also a net force.)

4. Gauss's law: field of an infinitely long line charge (λ)

Gaussian surface: coaxial cylinder of radius r, length l. Flux only through the curved surface: $\Phi = E\cdot 2\pi rl$.

Charge enclosed $= \lambda l$. $E\cdot 2\pi rl = \dfrac{\lambda l}{\varepsilon_0} \Rightarrow \boxed{E = \dfrac{\lambda}{2\pi\varepsilon_0 r}}$, radially outward (for $\lambda > 0$).

5. Gauss's law: infinite plane sheet (σ)

Gaussian surface: cylinder (pillbox) of cross-section A through the sheet. Flux only through the two flat ends: $2EA$.

Charge enclosed $= \sigma A$. $2EA = \dfrac{\sigma A}{\varepsilon_0} \Rightarrow \boxed{E = \dfrac{\sigma}{2\varepsilon_0}}$, independent of distance.

6. Gauss's law: thin spherical shell (charge q, radius R)

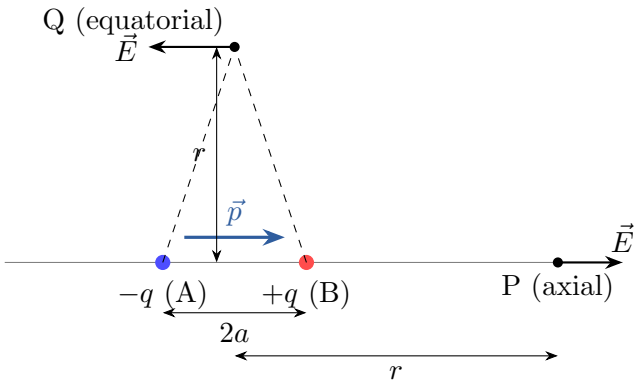
Gaussian sphere of radius r concentric with the shell: $\Phi = E\cdot 4\pi r^2$.

Outside $(r > R)$: encloses q $\Rightarrow \boxed{E = \dfrac{q}{4\pi\varepsilon_0 r^2}}$ (as if all charge at centre).

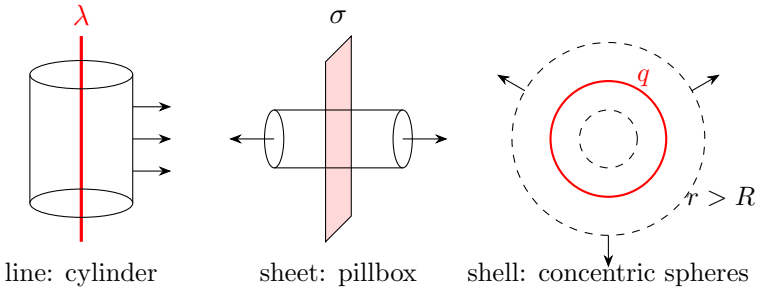
Inside $(r < R)$: encloses 0 $\Rightarrow \boxed{E = 0}$.

1B Diagrams to practise

Dipole: axial and equatorial points (labels: $-q$, $+q$, $2a$, r, P, $\vec{p}$, θ)



Gaussian surfaces for the three standard cases



Also practise (not drawn here): field lines of a single ± charge, of a dipole and of two equal positive charges (lines never cross; start on +, end on −; denser where E is stronger).

1C Exercise-wise numerical patterns

Exercises 1.1–1.23 (23 questions, 5 patterns)

Pattern A: Coulomb's law and superposition

Spot: force between point charges. **Method:** $F = \frac{kq_1q_2}{r^2}$; for several charges add vectors, use symmetry.

Solved (Q1.12): (a) $F = \dfrac{9\times10^9\times(6.5\times10^{-7})^2}{(0.50)^2} = 1.5\times10^{-2}$ N.

(b) Each charge doubled (×4), distance halved (×4): $F' = 16F = 0.24$ N.

Pattern B: Electric field and the dipole

Solved (Q1.8): at the midpoint each charge gives $E = \dfrac{9\times10^9\times3\times10^{-6}}{(0.10)^2} = 2.7\times10^6$ N/C, both pointing towards B (the negative charge).

Net $E = 5.4\times10^6$ N/C along OB. Force on -1.5×10^{-9} C: $F = qE = 8.1\times10^{-3}$ N along OA (opposite to $\vec{E}$).

Pattern C: Flux and Gauss's law ($\Phi = q_{\text{enc}}/\varepsilon_0$)

Solved (Q1.17): the square is one face of a cube of edge 10 cm with the charge at its centre, so it gets $\frac{1}{6}$ of the total flux.

$$\Phi = \frac{q}{6\varepsilon_0} = \frac{10 \times 10^{-6}}{6 \times 8.854 \times 10^{-12}} = 1.9 \times 10^{5}\ \text{N m}^2/\text{C}.$$

Pattern D: Gauss's law applications (sphere, line, sheet)

Solved (Q1.21): $r = 1.2$ m. $Q = 4\pi r^2\sigma = 4\pi(1.2)^2(80 \times 10^{-6}) = 1.45 \times 10^{-3}$ C.

$$\Phi = \frac{Q}{\varepsilon_0} = \frac{1.45 \times 10^{-3}}{8.854 \times 10^{-12}} = 1.6 \times 10^{8}\ \text{N m}^2/\text{C}.$$

Pattern E: Properties of charge (quantisation, conservation)

Solved (Q1.11): $n = \frac{q}{e} = \frac{3 \times 10^{-7}}{1.6 \times 10^{-19}} \approx 2 \times 10^{12}$ electrons, from wool to polythene.

Mass transferred $= n\,m_e \approx 2 \times 10^{-18}$ kg: yes, but negligible.

Pattern A: Q1.1, 1.2, 1.6, 1.12. Pattern B: Q1.8, 1.9, 1.10, 1.13. Pattern C: Q1.14–1.19. Pattern D: Q1.20–1.23. Pattern E: Q1.3, 1.4, 1.5, 1.7, 1.11.

1D Conceptual questions

- **Why do field lines never cross?** → There would be two directions of $\vec{E}$ at one point.
- **Why can't a field line have breaks?** → $\vec{E}$ is defined and continuous at every point in a charge-free region.
- **Why can quantisation be ignored for large charges?** → e is tiny compared with macroscopic charges; charge looks continuous.
- **Net flux through a closed surface is zero; are there no charges inside?** → Not necessarily; the *net* enclosed charge is zero.
- **Force on a dipole in a uniform field?** → Zero (only a torque); a non-uniform field gives a net force too.
- **Field inside a charged conductor / shell?** → Zero; charge resides on the outer surface.
- **Does the Gaussian surface depend on the charge outside?** → $\vec{E}$ on it does; the *net flux* does not.
- **Stable equilibrium of a dipole?** → $\vec{p}$ parallel to $\vec{E}$ ($\theta = 0$).

1E Exam tips

Common mistakes that lose marks

- Field direction: *away from* $+q$, *towards* $-q$. At the dipole midpoint both fields point towards $-q$.
- Units: $\mu\text{C} = 10^{-6}$ C, cm $\to$ m before squaring.
- Dipole moment points from $-q$ to $+q$.
- Using $E = \sigma/\varepsilon_0$ (conductor surface / between two plates) where $\sigma/2\varepsilon_0$ (single sheet) applies.
- Forgetting the vector nature in superposition; use symmetry to cancel.

Question types: 1-mark: field lines, flux, quantisation. 2–3 marks: Coulomb/superposition numericals, flux through a cube face. 5 marks: dipole field derivations, Gauss's law applications.

1F JEE extras

Useful results

- Dipole at general point: $E = \dfrac{p\sqrt{1+3\cos^2\theta}}{4\pi\varepsilon_0 r^3}$.
- Non-uniform field: $F = p\,\dfrac{dE}{dx}$ (dipole along x).
- Uniformly charged solid sphere: $E = \dfrac{kQr}{R^3}$ inside, $\dfrac{kQ}{r^2}$ outside.
- Charge at a cube corner: flux through the whole cube $= \dfrac{q}{8\varepsilon_0}$.

Example: q at a corner of a cube: 8 identical cubes would enclose it, so this cube gets $\frac{1}{8}$ of q/ε_0.

1G Pattern spotter

If you see...	Do this	See
Forces and fields		
Force between point charges	$F = kq_1q_2/r^2$; scale for changes in q, r	Ex A
Charges at corners of a square/triangle	Vector sum; use symmetry	Ex A
Field at a point due to charges	Add $\vec{E}$ vectors; direction away from $+$	Ex B
Dipole moment / torque	$p = q \cdot 2a$; $\tau = pE\sin\theta$	Ex B
Tracks of charged particles	Bends towards opposite plate; more bend $\Rightarrow$ larger q/m	Ex B
Flux and Gauss's law		
Flux of uniform field through a flat area	$\Phi = EA\cos\theta$	Ex C
Closed surface, charge inside	$\Phi = q/\varepsilon_0$ (shape doesn't matter)	Ex C
Charge at centre, one face asked	$\frac{1}{6}$ of q/ε_0	Ex C
Sphere / line / sheet	kQ/r^2; $\lambda/2\pi\varepsilon_0 r$; $\sigma/2\varepsilon_0$	Ex D

If you see...	Do this	See
Number of electrons transferred	$n = q/e$	Ex E

Chapter 2 Electrostatic Potential and Capacitance

2A Derivations (board favourites)

1. Potential due to a point charge

Work done to bring a unit positive charge from ∞ to distance r against $\vec{E}$:
$V = -\int_{\infty}^{r} \frac{q}{4\pi\varepsilon_0 r'^2} dr' = \boxed{\frac{q}{4\pi\varepsilon_0 r}}$.

2. Potential due to a dipole

At P (distance r, angle θ from $\vec{p}$): $V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$ with $r_{1,2} \approx r \mp a\cos\theta$ for $r \gg a$.

$\frac{1}{r_1} - \frac{1}{r_2} \approx \frac{2a\cos\theta}{r^2} \Rightarrow \boxed{V = \frac{p\cos\theta}{4\pi\varepsilon_0 r^2}}$. Axial: $\pm\frac{p}{4\pi\varepsilon_0 r^2}$; equatorial: 0.

3. Potential energy of a dipole in a uniform field

Work to rotate from θ_0 to θ_1: $W = \int_{\theta_0}^{\theta_1} pE\sin\theta\, d\theta = pE(\cos\theta_0 - \cos\theta_1)$.

Taking $\theta_0 = \frac{\pi}{2}$ as zero: $\boxed{U = -pE\cos\theta = -\vec{p}\cdot\vec{E}}$. Minimum $(-pE)$ at $\theta = 0$: stable.

4. Capacitance of a parallel plate capacitor (with dielectric)

Field between plates $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$; $V = Ed = \frac{Qd}{\varepsilon_0 A} \Rightarrow \boxed{C = \frac{\varepsilon_0 A}{d}}$.

With a dielectric filling the gap, E reduces by K: $C = \frac{K\varepsilon_0 A}{d}$. (Slab of thickness t: $C = \frac{\varepsilon_0 A}{d - t + t/K}$.)

5. Series and parallel combinations

Series: same Q; $V = V_1 + V_2 + \cdots = Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \ldots\right) \Rightarrow \boxed{\frac{1}{C} = \sum\frac{1}{C_i}}$.

Parallel: same V; $Q = Q_1 + Q_2 + \cdots = V(C_1 + C_2 + \ldots) \Rightarrow \boxed{C = \sum C_i}$.

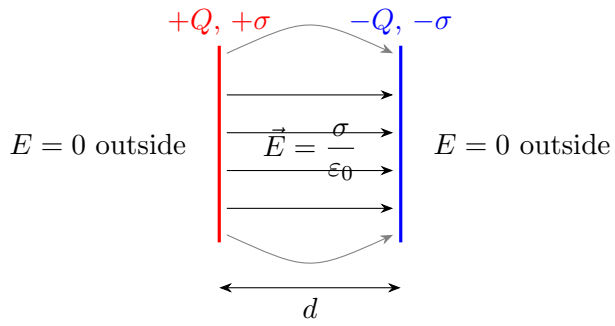
6. Energy stored in a capacitor; energy density

Bringing dq at potential q/C: $dW = \frac{q}{C}dq \Rightarrow W = \int_0^Q \frac{q}{C}dq = \boxed{\frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV}$.

Parallel plates: $U = \frac{1}{2}\frac{\varepsilon_0 A}{d}(Ed)^2 = \frac{1}{2}\varepsilon_0 E^2(Ad) \Rightarrow$ energy density $\boxed{u = \frac{1}{2}\varepsilon_0 E^2}$.

2B Diagrams to practise

Parallel plate capacitor (labels: $+\sigma$, $-\sigma$, d, A, uniform $\vec{E}$, fringing at edges)



Also practise: equipotential surfaces (concentric spheres for a point charge; planes for a uniform field; the dipole pattern), always drawn perpendicular to the field lines.

2C Exercise-wise numerical patterns

Exercises 2.1–2.11 (11 questions, 5 patterns)

Pattern A: Potential of a system of point charges

Method: $V = \sum \dfrac{kq_i}{r_i}$ (scalar, add with signs). For "$V = 0$" points, check between the charges *and* outside, beyond the smaller charge.

Solved (Q2.1): $q_1 = 5 \times 10^{-8}$ C at $x = 0$, $q_2 = -3 \times 10^{-8}$ C at $x = 16$ cm.
Between: $\frac{5}{x} = \frac{3}{16-x} \Rightarrow x = 10$ cm. Beyond q_2: $\frac{5}{x} = \frac{3}{x-16} \Rightarrow x = 40$ cm (from the positive charge).

Pattern B: Charged conductor (sphere)

Solved (Q2.4): (a) inside: $E = 0$. (b) just outside: $E = \dfrac{9 \times 10^9 \times 1.6 \times 10^{-7}}{(0.12)^2} = 10^5$ N/C.

(c) at 18 cm: $E = \dfrac{9 \times 10^9 \times 1.6 \times 10^{-7}}{(0.18)^2} = 4.4 \times 10^4$ N/C.

Pattern C: Capacitance and dielectrics (battery connected or not?)

Method: connected $\Rightarrow V$ fixed; disconnected $\Rightarrow Q$ fixed. Dielectric multiplies C by K.

Solved (Q2.8, 2.9): $C = \dfrac{\varepsilon_0 A}{d} = \dfrac{8.85 \times 10^{-12} \times 6 \times 10^{-3}}{3 \times 10^{-3}} = 1.8 \times 10^{-11}$ F $= 18$ pF; $Q = CV = 1.8 \times 10^{-9}$ C.
Mica ($K = 6$): $C' = 108$ pF. (a) connected: $V = 100$ V, $Q' = C'V = 1.08 \times 10^{-8}$ C. (b) disconnected: $Q = 1.8 \times 10^{-9}$ C, $V' = Q/C' = 16.7$ V.

Pattern D: Combinations of capacitors

Solved (Q2.6): three 9 pF in series: $\frac{1}{C} = \frac{3}{9} \Rightarrow C = 3$ pF. Same charge, equal capacitances, so the 120 V splits equally: 40 V each.

Pattern E: Energy stored and energy loss on sharing

Solved (Q2.11): before: $U_1 = \frac{1}{2}(600 \times 10^{-12})(200)^2 = 1.2 \times 10^{-5}$ J.
After joining an equal uncharged capacitor, charge is shared: $V = 100$ V on each: $U_2 = 2 \times \frac{1}{2}(600 \times 10^{-12})(100)^2 = 6 \times 10^{-6}$ J.
Energy lost $= 6 \times 10^{-6}$ J (as heat and radiation in the connecting wires).

Pattern A: Q2.1, 2.2, 2.3. Pattern B: Q2.4. Pattern C: Q2.5, 2.8, 2.9. Pattern D: Q2.6, 2.7. Pattern E: Q2.10, 2.11.

2D Conceptual questions

- **Why is $\vec{E}$ perpendicular to an equipotential surface?** → No work is done moving along it, so $\vec{E}$ has no component along it.
- **Why is the potential the same everywhere inside a conductor?** → $E = 0$ inside, so $dV = -E\,dr = 0$.
- **Why is E at a conductor's surface normal to it?** → A tangential component would move free charges.
- **Why do charges collect at sharp points?** → $\sigma \propto \frac{1}{R}$; high field there (lightning conductor, corona discharge).
- **What does a dielectric do?** → It gets polarised; its induced field opposes $\vec{E}$, so E falls and C rises by K.
- **Two equipotential surfaces cross?** → No; a point would have two potentials.
- **Work done moving a charge around a closed path in an electrostatic field?** → Zero (conservative field).

2E Exam tips

Common mistakes that lose marks

- Potential is a scalar: add with signs, no vector components.
- Mixing up "battery connected" (V constant) and "disconnected" (Q constant).
- pF $= 10^{-12}$ F, μF $= 10^{-6}$ F.
- Series rule for capacitors is the *parallel* rule for resistors (and vice versa).
- Forgetting that energy is lost when two capacitors share charge.

Question types: 1-mark: equipotentials, conductor properties, K effect. 2–3 marks: V of charge systems, combinations, energy. 5 marks: dipole potential, parallel plate capacitor with dielectric, energy density derivation.

2F JEE extras

Useful results

- $\vec{E} = -\nabla V$; $E = -\dfrac{dV}{dr}$.
- Common potential on sharing: $V = \dfrac{C_1V_1 + C_2V_2}{C_1 + C_2}$; energy lost $= \dfrac{C_1C_2(V_1 - V_2)^2}{2(C_1 + C_2)}$.
- Spherical capacitor: $C = 4\pi\varepsilon_0 \dfrac{ab}{b-a}$; isolated sphere: $C = 4\pi\varepsilon_0 R$.
- Force between plates: $F = \dfrac{Q^2}{2\varepsilon_0 A}$.

Example (Q2.11 again): lost $= \dfrac{600 \cdot 600 \times 10^{-24}(200)^2}{2 \times 1200 \times 10^{-12}} = 6 \times 10^{-6}$ J. ✓

2G Pattern spotter

If you see…	Do this	See
Potential		
Find V at a point	$\sum kq_i/r_i$ with signs	Ex A
Where is $V = 0$?	Check between charges and beyond the smaller one	Ex A
Equipotential surface of $+q$, $-q$	Perpendicular bisector plane	Ex A
Charged conducting sphere	$E = 0$ inside; kQ/r^2 outside	Ex B
Capacitors		
Change d or insert dielectric	$C = K\varepsilon_0 A/d$	Ex C
Battery stays connected	V fixed; Q changes	Ex C
Battery disconnected	Q fixed; V changes	Ex C
Series / parallel	$\frac{1}{C} = \sum \frac{1}{C_i}$ / $C = \sum C_i$	Ex D
Energy stored	$\frac{1}{2}CV^2 = \frac{Q^2}{2C}$	Ex E
Charged capacitor joined to another	Common V; energy is lost	Ex E

Chapter 3 Current Electricity

3A Derivations (board favourites)

1. Drift velocity and $I = neAv_d$

Acceleration of an electron $a = -\dfrac{eE}{m}$; average over relaxation time τ: $\boxed{v_d = -\dfrac{eE\tau}{m}}$.

In time Δt, charge crossing area A: $neA|v_d|\Delta t \Rightarrow \boxed{I = neAv_d}$.

2. Ohm's law and resistivity from drift

$j = \dfrac{I}{A} = nev_d = \dfrac{ne^2\tau}{m}E$, i.e. $j = \sigma E$ with $\boxed{\sigma = \dfrac{ne^2\tau}{m}}$, $\rho = \dfrac{m}{ne^2\tau}$.

Since $E = V/l$: $V = \dfrac{\rho l}{A}I = RI$ (Ohm's law). Mobility $\mu = \dfrac{v_d}{E} = \dfrac{e\tau}{m}$.

3. Internal resistance; cells in series and parallel

Terminal voltage $V = \varepsilon - Ir$, $I = \dfrac{\varepsilon}{R + r}$ (discharging); charging: $V = \varepsilon + Ir$.

Series: $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$, $r_{eq} = r_1 + r_2$.

Parallel: $\dfrac{\varepsilon_{eq}}{r_{eq}} = \dfrac{\varepsilon_1}{r_1} + \dfrac{\varepsilon_2}{r_2}$, $\dfrac{1}{r_{eq}} = \dfrac{1}{r_1} + \dfrac{1}{r_2}$.

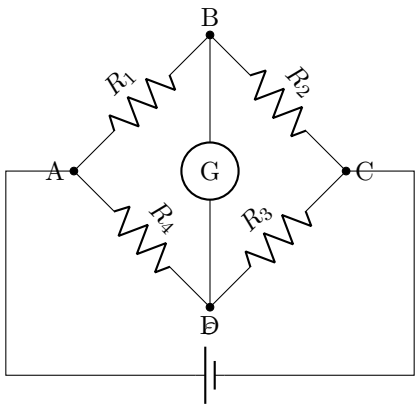
4. Wheatstone bridge balance condition

At balance $I_g = 0$, so B and D are at the same potential. Loop ABDA: $I_1R_1 = I_2R_4$; loop BCDB: $I_1R_2 = I_2R_3$.

Divide: $\boxed{\dfrac{R_1}{R_2} = \dfrac{R_4}{R_3}}$ (or $\frac{P}{Q} = \frac{R}{S}$ in the usual labelling).

3B Diagrams to practise

Wheatstone bridge (labels: R_1–R_4, G, I_1, I_2, I_g, cell, points A, B, C, D)



Also practise: I–V graph of an ohmic conductor (straight line) and of a non-ohmic device; ρ vs T for a metal (rises) and a semiconductor (falls).

3C Exercise-wise numerical patterns

Exercises 3.1–3.9 (9 questions, 4 patterns)

Pattern A: EMF, internal resistance, terminal voltage

Solved (Q3.8): charging current $I = \dfrac{120-8}{15.5+0.5} = 7$ A. Terminal voltage while charging $V = \varepsilon + Ir = 8 + 7\times 0.5 = 11.5$ V.
The series resistor limits the current; without it the current would be dangerously large.

Pattern B: Temperature dependence of resistance

Method: $R_T = R_0[1+\alpha(T-T_0)] \Rightarrow T - T_0 = \dfrac{R_T - R_0}{R_0\alpha}$.

Solved (Q3.6): $R_{27} = \frac{230}{3.2} = 71.9\ \Omega$, $R_T = \frac{230}{2.8} = 82.1\ \Omega$.

$$T - 27 = \frac{82.1 - 71.9}{71.9 \times 1.7 \times 10^{-4}} = 840 \Rightarrow T \approx 867\ ^\circ\text{C}.$$

Pattern C: Resistivity and drift speed

Solved (Q3.9): $v_d = \dfrac{I}{neA} = \dfrac{3}{8.5\times 10^{28}\times 1.6\times 10^{-19}\times 2\times 10^{-6}} = 1.1\times 10^{-4}$ m/s.

$$t = \frac{l}{v_d} = \frac{3}{1.1\times 10^{-4}} = 2.7\times 10^4\ \text{s} \approx 7.5\ \text{h}.$$

Pattern D: Kirchhoff's rules for a network

Method: mark currents using the junction rule, write one loop equation per independent loop, solve.

Solved (Q3.7): let the cell current be I, I_1 in AB, $I - I_1$ in AD, I_2 in BD.
Loop ABDA: $10I_1 + 5I_2 - 5(I - I_1) = 0$. Loop BCDB: $5(I_1 - I_2) - 10(I - I_1 + I_2) - 5I_2 = 0$. Outer loop: $10I + 10I_1 + 5(I_1 - I_2) = 10$.
Solving: $I = \frac{10}{17}$ A; AB $\frac{4}{17}$, BC $\frac{6}{17}$, AD $\frac{6}{17}$, DC $\frac{4}{17}$ A, and $\frac{2}{17}$ A flows from D to B.

Pattern A: Q3.1, 3.2, 3.8. Pattern B: Q3.3, 3.5, 3.6. Pattern C: Q3.4, 3.9. Pattern D: Q3.7.

3D Conceptual questions

- **Drift speed is tiny ($\sim 10^{-4}$ m/s); why does a bulb glow instantly?** $\rightarrow$ The field is set up along the whole wire almost at the speed of light.
- **Effect of temperature on resistivity of a metal?** $\rightarrow$ Increases: more collisions, smaller τ.
- **Of a semiconductor?** $\rightarrow$ Decreases: n increases sharply with T.
- **Why manganin/constantan for standard resistors?** $\rightarrow$ Very small temperature coefficient α, high ρ.
- **When is terminal voltage greater than EMF?** $\rightarrow$ While the cell is being charged.
- **Kirchhoff's junction rule is based on?** $\rightarrow$ Conservation of charge. Loop rule: conservation of energy.
- **Wheatstone bridge most sensitive when?** $\rightarrow$ All four resistances are of the same order.

3E Exam tips

Common mistakes that lose marks

- Sign convention in loops: going across a resistor *with* the current is $-IR$; from $-$ to $+$ inside a cell is $+\varepsilon$.
- Using $R_T = R_0(1 + \alpha\Delta T)$ with R_0 at the wrong reference temperature.
- Forgetting r (internal resistance) in $I = \varepsilon/(R + r)$.
- Negative current answer: it only means the actual direction is opposite to the one assumed.

Question types: 1-mark: drift speed, mobility, ρ–T graph. 2–3 marks: cells and internal resistance, Wheatstone bridge, temperature dependence. 5 marks: drift velocity derivation and $\rho = m/ne^2\tau$, Kirchhoff network.

3F JEE extras

Useful results

- Maximum power transfer: $R = r$, $P_{max} = \dfrac{\varepsilon^2}{4r}$.
- Stretching a wire n times (volume constant): $R' = n^2 R$.
- m rows of n identical cells: $I = \dfrac{n\varepsilon}{R + nr/m}$; maximum when $R = \dfrac{nr}{m}$.
- Balanced bridge: the galvanometer arm can be removed when simplifying a network.

Example: a 4 Ω wire stretched to twice its length: $R' = 2^2 \times 4 = 16\ \Omega$.

3G Pattern spotter

If you see...	Do this	See
Cells		
Maximum current from a battery	$I = \varepsilon/r$ (short circuit)	Ex A
Current, terminal voltage	$I = \varepsilon/(R+r)$; $V = \varepsilon - Ir$	Ex A
Battery being charged	$V = \varepsilon + Ir$	Ex A
Resistance		
Resistance at a new temperature	$R_T = R_0[1 + \alpha(T - T_0)]$	Ex B
Find α from two readings	$\alpha = \frac{R_2 - R_1}{R_1(T_2 - T_1)}$	Ex B
Resistivity from R, l, A	$\rho = RA/l$	Ex C
Time for an electron to drift	$v_d = I/neA$; $t = l/v_d$	Ex C
Networks		
Network with several loops	Kirchhoff's rules	Ex D
Bridge arrangement	Check $\frac{P}{Q} = \frac{R}{S}$ first	3A

Chapter 4 Moving Charges and Magnetism

4A Derivations (board favourites)

1. Field on the axis of a circular loop (Biot–Savart)

Element dl at distance $r = \sqrt{x^2 + R^2}$ from P: $dB = \dfrac{\mu_0}{4\pi}\dfrac{I\,dl}{x^2 + R^2}$ $(dl \perp r)$.
Perpendicular components cancel; axial components $dB\cos\theta$, $\cos\theta = \frac{R}{\sqrt{x^2+R^2}}$, add over $\oint dl = 2\pi R$:
$\boxed{B = \dfrac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}}$. At the centre $(x = 0)$: $B = \dfrac{\mu_0 I}{2R}$; N turns: multiply by N.

2. Ampere's law: long straight wire

Circular loop of radius r round the wire; B is tangential and constant on it: $\oint \vec{B}\cdot\vec{dl} = B\cdot 2\pi r = \mu_0 I$
$\Rightarrow \boxed{B = \dfrac{\mu_0 I}{2\pi r}}$ (right-hand thumb rule for direction).

3. Ampere's law: long solenoid

Rectangular loop abcd, side h inside along the axis. Only the inside side contributes: $Bh = \mu_0(nh)I$
$\Rightarrow \boxed{B = \mu_0 n I}$ inside (n = turns per unit length); $B \approx 0$ outside.

4. Force between two parallel currents; definition of ampere

Field of wire a at wire b: $B_a = \dfrac{\mu_0 I_a}{2\pi d}$. Force on length L of b: $F = I_b L B_a$.
$\boxed{\dfrac{F}{L} = \dfrac{\mu_0 I_a I_b}{2\pi d}}$; like currents attract, unlike repel.
1 ampere: the current which, in two long parallel wires 1 m apart in vacuum, gives a force of 2×10^{-7} N per metre.

5. Torque on a rectangular current loop

Sides a (perpendicular to $\vec{B}$) and b; normal makes angle θ with $\vec{B}$. Forces on the two sides of length a: $F = IaB$, opposite, separated by $b\sin\theta$.
$\tau = IaB\cdot b\sin\theta = IAB\sin\theta$; N turns: $\boxed{\vec{\tau} = \vec{m}\times\vec{B}}$, $m = NIA$.

6. Moving coil galvanometer

Radial field ($\theta = 90°$ always): deflecting torque $NIAB$ = restoring torque $k\phi$ $\Rightarrow \boxed{\phi = \dfrac{NAB}{k}I}$, so $\phi \propto I$.
Current sensitivity $\dfrac{\phi}{I} = \dfrac{NAB}{k}$; voltage sensitivity $\dfrac{\phi}{V} = \dfrac{NAB}{kR}$.
Ammeter: small shunt S in parallel, $S = \dfrac{I_g G}{I - I_g}$. **Voltmeter:** large R in series, $R = \dfrac{V}{I_g} - G$.

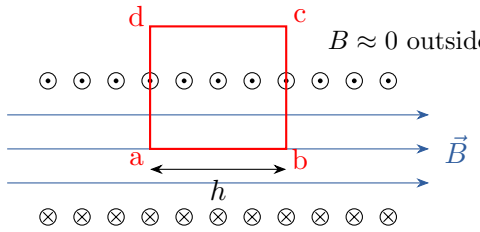
7. Charge in a uniform magnetic field

$\vec{v} \perp \vec{B}$: $qvB = \dfrac{mv^2}{r} \Rightarrow \boxed{r = \dfrac{mv}{qB}}$, $\omega = \dfrac{qB}{m}$, $\nu = \dfrac{qB}{2\pi m}$ (independent of v).

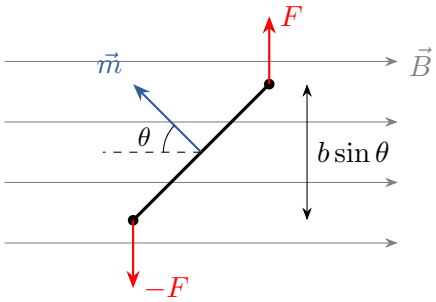
Velocity at angle to $\vec{B}$: helix, pitch $= v_{\parallel} \cdot \dfrac{2\pi m}{qB}$.

4B Diagrams to practise

Ampere loop for a solenoid (labels: rectangle abcd, h, $\vec{B}$ inside, current in/out)



Torque on a rectangular coil (top view; labels: $\vec{B}$, $\vec{m}$ (normal), θ, F, $-F$, $b\sin\theta$)



Also practise: field lines of a straight wire (concentric circles), of a circular loop and of a solenoid; the moving coil galvanometer (coil, radial field from concave poles, soft-iron core, spring, pointer).

4C Exercise-wise numerical patterns

Exercises 4.1–4.13 (13 questions, 4 patterns)

Pattern A: Field due to a current (wire, coil, solenoid)

Solved (Q4.8): total turns $= 5 \times 400 = 2000$ on 0.8 m: $n = 2500\ \text{m}^{-1}$.
$B = \mu_0 n I = 4\pi \times 10^{-7} \times 2500 \times 8.0 = 2.5 \times 10^{-2}$ T (the diameter is not needed).

Pattern B: Force on a conductor / between parallel wires

Solved (Q4.7): $F = \dfrac{\mu_0 I_A I_B L}{2\pi d} = \dfrac{2 \times 10^{-7} \times 8 \times 5 \times 0.10}{0.04} = 2 \times 10^{-5}$ N, attractive (same direction currents), towards B.

Pattern C: Torque on a coil; galvanometer sensitivity

Solved (Q4.9): $\tau = NIAB \sin\theta = 20 \times 12 \times (0.1)^2 \times 0.80 \times \sin 30^\circ = 0.96$ N m.
(Q4.10): current sensitivity ratio $= \dfrac{N_2 A_2 B_2}{N_1 A_1 B_1} = \dfrac{42 \times 1.8 \times 0.50}{30 \times 3.6 \times 0.25} = 1.4$; voltage sensitivity ratio $= 1.4 \times \dfrac{R_1}{R_2} = 1.4 \times \dfrac{10}{14} = 1$.

Pattern D: Charged particle in a magnetic field

Solved (Q4.11, 4.12): force $q\vec{v} \times \vec{B}$ is always $\perp \vec{v}$, so it provides centripetal force: circular path.
$r = \dfrac{mv}{eB} = \dfrac{9.1 \times 10^{-31} \times 4.8 \times 10^{6}}{1.6 \times 10^{-19} \times 6.5 \times 10^{-4}} = 4.2$ cm. $\nu = \dfrac{eB}{2\pi m} = 18$ MHz, independent of speed.
(The question prints $e = 1.5 \times 10^{-19}$ C, which gives 4.5 cm; NCERT's answer uses 1.6×10^{-19} C.)

Pattern A: Q4.1, 4.2, 4.3, 4.4, 4.8. Pattern B: Q4.5, 4.6, 4.7. Pattern C: Q4.9, 4.10, 4.13. Pattern D: Q4.11, 4.12.

4D Conceptual questions

- **Can a magnetic field change a charge's speed?** → No; force $\perp \vec{v}$, so no work is done; only direction changes.
- **Why a radial field in a galvanometer?** → Plane of the coil always parallel to $\vec{B}$, so $\tau = NIAB$ and $\phi \propto I$ (linear scale).
- **Why a soft-iron core?** → Increases B (sensitivity) and makes the field radial.
- **Does the torque depend on the coil's shape?** → No; only on area: $\tau = NIAB\sin\theta$ for any planar loop.
- **Ammeter in series with low resistance, voltmeter in parallel with high resistance: why?** → So that they don't change the current/voltage being measured.
- **Increasing current sensitivity always increases voltage sensitivity?** → No; increasing N also increases R.

4E Exam tips

Common mistakes that lose marks

- Direction rules: right-hand thumb rule (field of a wire), Fleming's left-hand rule (force), $\vec{F} = q\vec{v}\times\vec{B}$ (reverse for electrons).
- θ in $\tau = NIAB\sin\theta$ is between the *normal* and $\vec{B}$, not the plane and $\vec{B}$.
- Gauss (G) to tesla: 1 G $= 10^{-4}$ T.
- n in $\mu_0 nI$ is turns *per metre*, counting all layers.

Question types: 1-mark: direction of force/field, cyclotron frequency independence. 2–3 marks: field numericals, force between wires, galvanometer conversion. 5 marks: Biot–Savart on the axis of a loop, Ampere for solenoid, torque on loop and MCG.

4F JEE extras

Useful results

- Finite wire: $B = \dfrac{\mu_0 I}{4\pi d}(\sin\alpha_1 + \sin\alpha_2)$.
- Arc of angle θ at its centre: $B = \dfrac{\mu_0 I\theta}{4\pi R}$.
- Toroid: $B = \dfrac{\mu_0 NI}{2\pi r}$ inside, zero outside.
- Orbiting electron: $m = \dfrac{evr}{2}$; Bohr magneton $\mu_B = \dfrac{eh}{4\pi m_e} = 9.27\times10^{-24}$ A m^2.

Example: semicircular wire of radius R: $B = \dfrac{\mu_0 I\pi}{4\pi R} = \dfrac{\mu_0 I}{4R}$.

4G Pattern spotter

If you see...	Do this	See
Field due to currents		
Centre of a coil	$\mu_0 NI/2R$	Ex A
Near a long straight wire	$\mu_0 I/2\pi r$; direction by right-hand rule	Ex A
Inside a solenoid	$\mu_0 nI$, n per metre	Ex A
Forces and torques		
Wire in a field at an angle	$F = IlB\sin\theta$	Ex B
Two parallel wires	$F/L = \mu_0 I_1 I_2/2\pi d$	Ex B
Coil in a field	$\tau = NIAB\sin\theta$ (angle with normal)	Ex C
Compare galvanometers	NAB/k and NAB/kR	Ex C
Charged particles		
Charge moving $\perp \vec{B}$	$r = mv/qB$; $\nu = qB/2\pi m$	Ex D

If you see…	Do this	See
Galvanometer to ammeter / voltmeter	Shunt S / series R	4A

Chapter 5 Magnetism and Matter

5A Derivations (board favourites)

1. Bar magnet as an equivalent solenoid

A solenoid of N turns, area A, current I has magnetic moment $m = NIA$; its field far away on the axis is $B = \dfrac{\mu_0}{4\pi}\dfrac{2m}{r^3}$, exactly like a bar magnet. So a bar magnet behaves like a current-carrying solenoid.

2. Torque and potential energy of a magnetic dipole in a uniform field

Forces $+q_m B$ and $-q_m B$ form a couple: $\boxed{\vec{\tau} = \vec{m} \times \vec{B}}$, $\tau = mB\sin\theta$.

$U = \displaystyle\int \tau\, d\theta = \boxed{-mB\cos\theta = -\vec{m}\cdot\vec{B}}$. Stable: $\theta = 0$ ($U = -mB$); unstable: $\theta = 180^\circ$ ($U = +mB$).

Work to rotate from θ_1 to θ_2: $W = mB(\cos\theta_1 - \cos\theta_2)$.

3. Field of a bar magnet (by analogy with the electric dipole)

Replace $p \to m$, $\frac{1}{\varepsilon_0} \to \mu_0$: axial $\boxed{B = \dfrac{\mu_0}{4\pi}\dfrac{2m}{r^3}}$ (along $\vec{m}$); equatorial $\boxed{B = \dfrac{\mu_0}{4\pi}\dfrac{m}{r^3}}$ (opposite to $\vec{m}$).

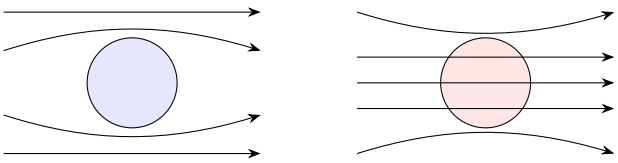
4. Magnetisation, intensity, susceptibility

$\vec{M} = \dfrac{\vec{m}_{net}}{V}$; $\vec{B} = \mu_0(\vec{H} + \vec{M})$; $M = \chi H \Rightarrow B = \mu_0(1+\chi)H = \mu_0\mu_r H = \mu H$, $\mu_r = 1 + \chi$.

5B Diagrams to practise

Practise: field lines of a bar magnet (closed loops, outside N→S, inside S→N); field lines *in* a diamagnetic sample (pushed out) and a paramagnetic/ferromagnetic sample (drawn in, concentrated).

Behaviour of materials in a magnetic field



diamagnetic: lines pushed outpara/ferro: lines drawn in

5C Exercise-wise numerical patterns

Exercises 5.1–5.7 (7 questions, 3 patterns)

Pattern A: Torque and energy of a magnetic dipole

Solved (Q5.5): work $W = mB(\cos 0^\circ - \cos\theta)$, $mB = 1.5 \times 0.22 = 0.33$ J.
(i) to 90°: $W = 0.33$ J; (ii) to 180°: $W = 0.66$ J. Torque: (i) $mB\sin 90^\circ = 0.33$ N m; (ii) $mB\sin 180^\circ = 0$.

Pattern B: Solenoid as a magnet ($m = NIA$)

Solved (Q5.3, 5.4): $m = NIA = 800 \times 3.0 \times 2.5 \times 10^{-4} = 0.60$ J T^{-1}, along the axis (right-hand rule).
Torque at 30° in 0.25 T: $\tau = mB\sin\theta = 0.60 \times 0.25 \times 0.5 = 7.5 \times 10^{-2}$ N m.
(Q5.4 says "Exercise 5.5"; it means the solenoid of Q5.3.)

Pattern C: Field of a short bar magnet

Solved (Q5.7): axial: $B = \dfrac{\mu_0}{4\pi}\dfrac{2m}{r^3} = 10^{-7} \times \dfrac{2 \times 0.48}{(0.1)^3} = 9.6 \times 10^{-5}$ T $= 0.96$ G (S→N, along $\vec{m}$).
Equatorial: half of this, 0.48 G, opposite to $\vec{m}$ (N→S).

Pattern A: Q5.1, 5.2, 5.5. Pattern B: Q5.3, 5.4, 5.6. Pattern C: Q5.7.

5D Conceptual questions

- **Why don't magnetic monopoles exist (Gauss's law for magnetism)?** → Net magnetic flux through any closed surface is zero: $\oint \vec{B} \cdot d\vec{S} = 0$.
- **Force on a magnet in a uniform field?** → Zero; only a torque.
- **Diamagnetic substance in a non-uniform field?** → Moves towards the weaker field; χ small and negative.
- **Paramagnetic?** → Weakly attracted towards the stronger field; χ small and positive.
- **Ferromagnetic?** → Strongly attracted; χ very large; domains align.
- **Superconductor?** → Perfect diamagnet ($\chi = -1$): Meissner effect, field expelled.
- **Effect of temperature?** → Diamagnetism unaffected; paramagnetism decreases; ferro becomes para above the Curie temperature.

5E Exam tips

Common mistakes that lose marks

- Work done is the *change* in U: $W = U_f - U_i$, not just $-mB\cos\theta$.
- Torque is maximum at 90°, zero at 0° and 180°.
- Axial field is *twice* the equatorial field at the same distance.
- Units: m in J T^{-1} or A m^2; 1 G $= 10^{-4}$ T.

Question types: 1-mark: classify a material by χ or μ_r, field-line behaviour. 2–3 marks: torque/energy numericals, comparison of dia/para/ferro. 5 marks: bar magnet as solenoid, properties table.

5F JEE extras

Useful results

- Curie's law (paramagnets): $\chi = \dfrac{C\mu_0}{T}$. Curie–Weiss (ferro above T_c): $\chi = \dfrac{C}{T - T_c}$.
- Hysteresis: retentivity and coercivity; permanent magnets need high both (steel); electromagnet cores need low coercivity (soft iron).
- Time period of a magnet oscillating in a field: $T = 2\pi\sqrt{\dfrac{I}{mB}}$.

Example: doubling B reduces T by a factor $\sqrt{2}$.

5G Pattern spotter

If you see...	Do this	See
Dipole in a field		
Torque on a magnet	$\tau = mB\sin\theta$	Ex A
Stable / unstable equilibrium	$\theta = 0$ ($U = -mB$) / 180° ($U = +mB$)	Ex A
Work to rotate	$W = mB(\cos\theta_1 - \cos\theta_2)$	Ex A
Solenoid's magnetic moment	$m = NIA$	Ex B
Field of a magnet		
Axial point	$\frac{\mu_0}{4\pi}\frac{2m}{r^3}$	Ex C
Equatorial point	$\frac{\mu_0}{4\pi}\frac{m}{r^3}$, opposite to $\vec{m}$	Ex C
Materials		
$\chi < 0$ / small > 0 / very large	Dia / para / ferro	5D

Chapter 6 Electromagnetic Induction

6A Derivations (board favourites)

1. Motional emf, $\varepsilon = Blv$

Rod of length l moving with speed v on rails, $\vec{B}$ perpendicular. Flux $\Phi = Blx$; $\dfrac{dx}{dt} = -v$.
$\varepsilon = -\dfrac{d\Phi}{dt} = \boxed{Blv}$. (Lorentz-force view: force qvB on charges along the rod; work per unit charge $= vBl$.)

2. Rod rotating about one end

Element dr at radius r moves at ωr: $d\varepsilon = B\omega r\, dr \;\Rightarrow\; \boxed{\varepsilon = \tfrac{1}{2}B\omega l^2}$.

3. Self-inductance of a long solenoid

$B = \mu_0 nI$; flux linkage $N\Phi = (nl)(\mu_0 nI)A \;\Rightarrow\; \boxed{L = \mu_0 n^2 Al}$ ($\mu_r\mu_0 n^2 Al$ with a core).

4. Mutual inductance of two coaxial solenoids

Inner S_1 (radius r_1, n_1), outer S_2 (n_2), common length l. Field of S_2: $\mu_0 n_2 I_2$; linkage with S_1: $(n_1 l)(\pi r_1^2)\mu_0 n_2 I_2$.
$\boxed{M_{12} = M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l}$.

5. Energy stored in an inductor

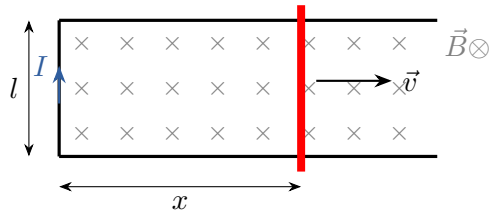
Work against back emf: $dW = LI\, dI \;\Rightarrow\; \boxed{W = \tfrac{1}{2}LI^2}$.

6. AC generator

Coil of N turns, area A, rotating at ω: $\Phi_B = BA\cos\omega t$.
$\varepsilon = -N\dfrac{d\Phi_B}{dt} = NBA\omega\sin\omega t \;\Rightarrow\; \boxed{\varepsilon = \varepsilon_0\sin\omega t}$, $\varepsilon_0 = NBA\omega$.

6B Diagrams to practise

Motional emf: rod on rails (labels: $\vec{B}$ into page, l, $\vec{v}$, I, x)



Also practise: AC generator (coil, magnetic poles, slip rings, brushes); Lenz's law sketches showing magnet approach/recede and the induced current direction.

6C Exercise-wise numerical patterns

Exercises 6.1–6.8 (8 questions, 3 patterns)

Pattern A: Direction of induced current (Lenz's law)

Method: is the flux through the loop increasing or decreasing? The induced current makes a field that opposes the *change*; then use the right-hand rule.

Solved (Q6.2(a)): an irregular wire turning into a circle increases its area, so flux increases; the induced current flows so as to reduce it, i.e. its field opposes $\vec{B}$ inside the loop.

Pattern B: Faraday's law with changing current; self and mutual inductance

Solved (Q6.3): $n = 15$ per cm $= 1500$ m^{-1}. $\dfrac{dB}{dt} = \mu_0 n \dfrac{dI}{dt} = 4\pi \times 10^{-7} \times 1500 \times \dfrac{2.0}{0.1} = 3.77 \times 10^{-2}$ T/s.

$\varepsilon = A\dfrac{dB}{dt} = 2.0 \times 10^{-4} \times 3.77 \times 10^{-2} = 7.5 \times 10^{-6}$ V.

(Q6.7): $L = \dfrac{\varepsilon}{dI/dt} = \dfrac{200}{5/0.1} = 4$ H. **(Q6.8):** $\Delta\Phi = M\Delta I = 1.5 \times 20 = 30$ Wb.

Pattern C: Motional emf (moving or rotating conductor)

Solved (Q6.5): $\varepsilon = \frac{1}{2}B\omega l^2 = \frac{1}{2} \times 0.5 \times 400 \times (1.0)^2 = 100$ V.

(Q6.4): $\varepsilon = Blv$ with l the side *perpendicular* to $\vec{v}$ (the side that cuts the field).

(a) v normal to the longer side: $\varepsilon = 0.3 \times 0.08 \times 0.01 = 2.4 \times 10^{-4}$ V for $\frac{0.02}{0.01} = 2$ s. (b) 6×10^{-5} V for 8 s.

Pattern A: Q6.1, 6.2. Pattern B: Q6.3, 6.7, 6.8. Pattern C: Q6.4, 6.5, 6.6.

6D Conceptual questions

- **Lenz's law follows from?** → Conservation of energy (otherwise current would grow without work being done).
- **Magnet dropped through a copper tube falls slowly: why?** → Induced currents oppose its motion.
- **Why is inductance called electrical inertia?** → It opposes the *change* of current, as mass opposes change of velocity.
- **Factors for mutual inductance?** → Number of turns, geometry, separation and orientation, and the core material.
- **Coil plane parallel to $\vec{B}$: induced emf in a moving loop?** → Zero flux, so zero emf (if flux doesn't change).
- **Which end of a falling E–W wire is at higher potential (Q6.6)?** → East end (emf from west to east).

6E Exam tips

Common mistakes that lose marks

- In $\varepsilon = Blv$, l is the length *perpendicular* to both $\vec{v}$ and $\vec{B}$.
- Turns per cm → per metre ($\times 100$).
- The minus sign in Faraday's law is Lenz's law; give the magnitude unless asked for direction.
- Rotating rod: $\frac{1}{2}B\omega l^2$, not $B\omega l^2$.

Question types: 1-mark: Lenz direction, units of L and M. 2–3 marks: $\varepsilon = Blv$, rotating rod, L from ε. 5 marks: self/mutual inductance of solenoids, energy in inductor, AC generator.

6F JEE extras

Useful results

- Charge through a coil: $q = \dfrac{N\Delta\Phi}{R}$ (independent of time taken).
- Rod on rails with resistance R: $I = \dfrac{Blv}{R}$, retarding force $\dfrac{B^2l^2v}{R}$, power $\dfrac{B^2l^2v^2}{R}$.
- Coupling: $M = k\sqrt{L_1L_2}$, $0 \le k \le 1$.
- Energy density of magnetic field: $u_B = \dfrac{B^2}{2\mu_0}$.

Example: coil of 100 turns, $R = 5\ \Omega$, flux per turn changes by 10^{-3} Wb: $q = \frac{100\times10^{-3}}{5} = 0.02$ C.

6G Pattern spotter

If you see...	Do this	See
Direction		
Find direction of induced current	Flux increasing or decreasing? Oppose the change	Ex A
Magnitude		
Changing current in a solenoid, loop inside	$\varepsilon = A\mu_0 n\, dI/dt$	Ex B
Given ε and dI/dt	$L = \varepsilon/(dI/dt)$	Ex B
Mutual inductance, change of flux	$\Delta\Phi = M\Delta I$	Ex B
Straight conductor moving	$\varepsilon = Blv$	Ex C
Rod rotating about one end	$\frac{1}{2}B\omega l^2$	Ex C
Loop leaving a field region	$\varepsilon = Blv$ for time (distance)$/v$	Ex C

Chapter 7 Alternating Current

7A Derivations (board favourites)

1. rms value of current

$i = i_m \sin\omega t$; power in R: $p = i^2R$; average of $\sin^2\omega t$ over a cycle $= \frac{1}{2}$.

$\bar{p} = \frac{1}{2} i_m^2 R = I^2 R \Rightarrow \boxed{I = \frac{i_m}{\sqrt{2}} = 0.707\, i_m}$; similarly $V = \frac{v_m}{\sqrt{2}}$.

2. AC through a pure inductor

$v_m \sin\omega t = L\frac{di}{dt} \Rightarrow i = -\frac{v_m}{\omega L}\cos\omega t = i_m \sin\left(\omega t - \frac{\pi}{2}\right)$.

$\boxed{X_L = \omega L}$; current **lags** voltage by $\frac{\pi}{2}$; average power $= 0$.

3. AC through a pure capacitor

$q = Cv_m \sin\omega t \Rightarrow i = \frac{dq}{dt} = \omega C v_m \cos\omega t = i_m \sin\left(\omega t + \frac{\pi}{2}\right)$.

$\boxed{X_C = \frac{1}{\omega C}}$; current **leads** voltage by $\frac{\pi}{2}$; average power $= 0$.

4. Series LCR: impedance by phasors

$\vec{V}_R$ along $\vec{I}$; $\vec{V}_L$ ahead by 90°; $\vec{V}_C$ behind by 90°. $v_m^2 = v_{Rm}^2 + (v_{Cm} - v_{Lm})^2$.

$\boxed{Z = \sqrt{R^2 + (X_C - X_L)^2}}$, $i_m = \frac{v_m}{Z}$, $\tan\phi = \frac{X_C - X_L}{R}$.

5. Resonance

Z is minimum $(= R)$ when $X_L = X_C$: $\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$; current is maximum, V_L and V_C cancel.

Quality factor $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$.

6. Average power and power factor

$p = vi = v_m i_m \sin\omega t \sin(\omega t + \phi)$; average: $\boxed{P = VI\cos\phi}$, power factor $\cos\phi = \frac{R}{Z}$.

Pure L or C: $\phi = \pm\frac{\pi}{2}$, $P = 0$ (wattless current). Resonance: $\cos\phi = 1$, $P = VI$.

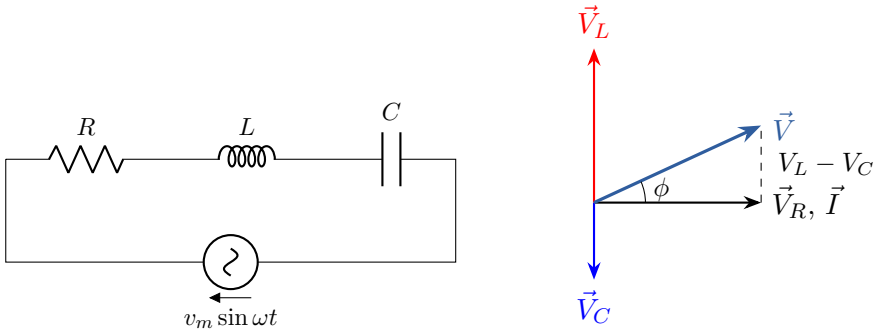
7. Transformer

Same flux change per turn: $\varepsilon_s = -N_s\frac{d\Phi}{dt}$, $\varepsilon_p = -N_p\frac{d\Phi}{dt} \Rightarrow \boxed{\frac{V_s}{V_p} = \frac{N_s}{N_p}}$; ideal $(V_pI_p = V_sI_s)$: $\frac{I_s}{I_p} = \frac{N_p}{N_s}$.

Losses: flux leakage, copper (I^2R), eddy currents (laminated core), hysteresis.

7B Diagrams to practise

Series LCR circuit and its phasor diagram (labels: $\vec{V}_R$, $\vec{V}_L$, $\vec{V}_C$, $\vec{V}$, $\vec{I}$, ϕ)



Also practise: resonance curve (i_m vs ω, sharper for smaller R); phasors for pure R, L, C; step-up transformer (laminated core, primary and secondary windings).

7C Exercise-wise numerical patterns

Exercises 7.1–7.8 (8 questions, 3 patterns)

Pattern A: Pure R, L or C; rms and peak values

Solved (Q7.4): $X_C = \dfrac{1}{2\pi f C} = \dfrac{1}{2\pi \times 60 \times 60 \times 10^{-6}} = 44.2\ \Omega$; $I = \dfrac{110}{44.2} = 2.49$ A.
(Q7.5): average power in a pure L or C is zero: $\phi = \pm 90°$, $\cos\phi = 0$.

Pattern B: LC circuit / natural frequency

Solved (Q7.6): $\omega = \dfrac{1}{\sqrt{LC}} = \dfrac{1}{\sqrt{27 \times 10^{-3} \times 30 \times 10^{-6}}} = 1.1 \times 10^3$ rad/s.

Pattern C: Series LCR at resonance

Solved (Q7.8): (a) $\omega_0 = \dfrac{1}{\sqrt{5.0 \times 80 \times 10^{-6}}} = 50$ rad/s, $f_0 = \dfrac{50}{2\pi} = 7.96$ Hz.
(b) $Z = R = 40\ \Omega$; $I_{rms} = \frac{230}{40} = 5.75$ A; amplitude $= 5.75\sqrt{2} = 8.13$ A.
(c) $V_R = 5.75 \times 40 = 230$ V; $V_L = I\omega_0 L = 5.75 \times 50 \times 5 = 1437.5$ V; $V_C = \dfrac{I}{\omega_0 C} = 1437.5$ V. V_L and V_C are 180° apart, so the LC drop is 0.

Pattern A: Q7.1, 7.2, 7.3, 7.4, 7.5. Pattern B: Q7.6. Pattern C: Q7.7 ($P = V^2/R = 2000$ W), 7.8.

7D Conceptual questions

- **Why is AC preferred for transmission?** → Voltage can be stepped up with transformers; high V, low I, small I^2R loss.
- **Capacitor blocks DC but passes AC: why?** → $X_C = 1/\omega C$ is infinite for $\omega = 0$.
- **What is wattless current?** → Current component $\perp$ to voltage ($I \sin\phi$); it does no work.
- **Why a laminated core?** → To reduce eddy-current losses.
- **Low power factor causes large loss: why?** → For the same P, $I = P/V\cos\phi$ is larger, so I^2R loss rises. A capacitor in parallel improves it.
- **At resonance, V_L can exceed the supply voltage: is that a violation?** → No; V_L and V_C are opposite in phase and cancel.

7E Exam tips

Common mistakes that lose marks

- Use $\omega = 2\pi f$; $X_L = 2\pi f L$, not fL.
- Meters read rms values; "peak" or "amplitude" means $\times\sqrt{2}$.
- Phase: inductor current lags (ELI), capacitor current leads (ICE).
- Add voltages as phasors, not as numbers ($V \neq V_R + V_L + V_C$).
- mH, μF conversions.

Question types: 1-mark: phase relations, wattless current, rms. 2–3 marks: reactance and current numericals, resonance frequency, transformer ratio. 5 marks: LCR impedance with phasor diagram, resonance, power factor, transformer working and losses.

7F JEE extras

Useful results

- Bandwidth $\Delta\omega = \dfrac{R}{L}$; $Q = \dfrac{\omega_0}{\Delta\omega}$.
- LC oscillations: $q = q_0\cos\omega_0 t$; energy shuttles between $\frac{q^2}{2C}$ and $\frac{1}{2}Li^2$.
- Average value over a half cycle: $\bar{i} = \dfrac{2i_m}{\pi} = 0.637\, i_m$.
- Parallel LC at resonance: impedance maximum (rejector circuit).

Example (Q7.8): $Q = \dfrac{\omega_0 L}{R} = \dfrac{50 \times 5}{40} = 6.25$.

7G Pattern spotter

If you see...	Do this	See
Single elements		
rms from peak / peak from rms	divide / multiply by $\sqrt{2}$	Ex A
Current through R, L or C	$I = V/R$, V/X_L, V/X_C	Ex A
Power in pure L or C	Zero	Ex A
Combinations		
L and C only, natural frequency	$\omega = 1/\sqrt{LC}$	Ex B
LCR at resonance	$Z = R$, $\omega_0 = 1/\sqrt{LC}$, max current	Ex C
Voltages across elements	IR, IX_L, IX_C; LC drop zero at resonance	Ex C
Average power	$P = VI\cos\phi = I^2R$	Ex C
Transformer	$V_s/V_p = N_s/N_p$	7A

Chapter 8 Electromagnetic Waves

8A Derivations (board favourites)

1. Displacement current

While a capacitor charges, Ampere's law gives different answers for a surface through the wire and one between the plates (no conduction current). Maxwell added:

$\Phi_E = EA = \dfrac{q}{\varepsilon_0} \Rightarrow \varepsilon_0 \dfrac{d\Phi_E}{dt} = \dfrac{dq}{dt} = i;$ $\boxed{i_d = \varepsilon_0 \dfrac{d\Phi_E}{dt}}$.

Ampere–Maxwell law: $\oint \vec{B} \cdot d\vec{l} = \mu_0(i_c + i_d)$. Between the plates $i_d = i_c$ in the wire.

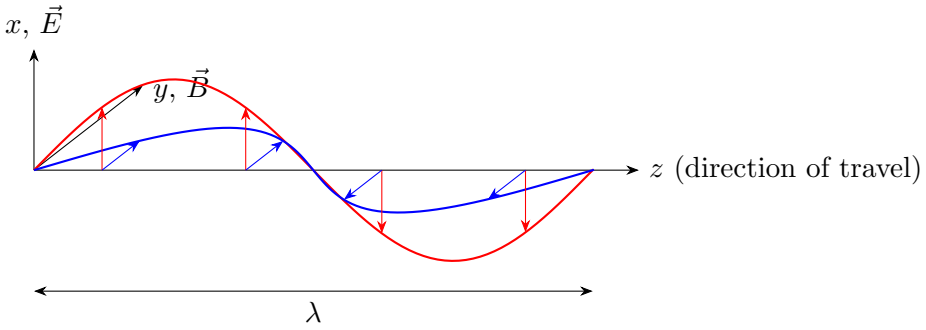
2. Properties of EM waves

$\vec{E} \perp \vec{B} \perp$ direction of travel; transverse; $E_0 = cB_0$; speed $\boxed{c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}}} = 3 \times 10^8$ m/s; in a medium $v = \dfrac{1}{\sqrt{\mu\varepsilon}}$.

$E_x = E_0 \sin(kz - \omega t)$, $B_y = B_0 \sin(kz - \omega t)$, $k = \dfrac{2\pi}{\lambda}$, $\omega = ck$. Produced by *accelerated* charges; no medium needed.

8B Diagrams to practise

Plane EM wave (labels: $\vec{E}$ along x, $\vec{B}$ along y, travel along z, λ)



8C Exercise-wise numerical patterns

Exercises 8.1–8.10 (10 questions, 3 patterns)

Pattern A: Displacement current in a capacitor

Solved (Q8.1): $C = \dfrac{\varepsilon_0 \pi r^2}{d} = \dfrac{8.85 \times 10^{-12} \times \pi (0.12)^2}{0.05} = 8.0 \times 10^{-12}$ F.
$\dfrac{dV}{dt} = \dfrac{I}{C} = \dfrac{0.15}{8.0 \times 10^{-12}} = 1.87 \times 10^{10}$ V/s. $i_d = i_c = 0.15$ A. Kirchhoff's junction rule holds if the displacement current is included.

Pattern B: Wave relations ($c = \nu\lambda$, $E_0 = cB_0$, ω, k)

Solved (Q8.8): $B_0 = \dfrac{E_0}{c} = \dfrac{120}{3 \times 10^8} = 4 \times 10^{-7}$ T; $\omega = 2\pi\nu = 3.14 \times 10^8$ rad/s; $k = \dfrac{\omega}{c} = 1.05$ rad/m; $\lambda = \dfrac{c}{\nu} = 6.0$ m.
$\vec{E} = 120 \sin(1.05x - 3.14 \times 10^8 t)\,\hat{j}$ N/C, $\vec{B} = 4 \times 10^{-7} \sin(1.05x - 3.14 \times 10^8 t)\,\hat{k}$ T (one valid choice of axes).

Pattern C: Spectrum and photon energy

Method: $E = h\nu = \dfrac{hc}{\lambda}$; in eV: $E(\text{eV}) \approx \dfrac{1240}{\lambda(\text{nm})}$.

Solved (Q8.3): X-rays, red light and radio waves all travel at the same speed c in vacuum.

Pattern A: Q8.1, 8.2. Pattern B: Q8.4, 8.5, 8.6, 8.7, 8.8, 8.10. Pattern C: Q8.3, 8.9.

8D Conceptual questions and the spectrum

Wave	λ (approx.)	Produced by	Uses
Radio	> 0.1 m	Accelerating electrons in aerials	Radio, TV, mobile phones
Microwave	1 mm – 0.1 m	Klystron or magnetron valve	Radar, microwave ovens
Infrared	700 nm – 1 mm	Vibrating atoms and molecules	Remote controls, greenhouse effect, physiotherapy
Visible	400 – 700 nm	Electrons changing energy levels	Vision, photography
Ultraviolet	1 – 400 nm	Inner-shell electron transitions; the Sun	Sterilising (water purifiers), LASIK
X-rays	10^{-3} – 1 nm	X-ray tubes, inner-shell electrons	Medical imaging, cancer treatment
Gamma	$< 10^{-3}$ nm	Radioactive decay of nuclei	Cancer treatment, nuclear studies

- **Why is displacement current needed?** → To make Ampere's law consistent while the electric flux changes (charging capacitor).

- **Do EM waves carry energy and momentum?** → Yes; they exert radiation pressure.
- **What is the frequency of EM waves from a charge oscillating at ν?** → The same, ν (Q8.6).
- **Why is the ozone layer important?** → It absorbs harmful UV.

8E Exam tips

Common mistakes that lose marks

- Order of the spectrum (increasing λ): gamma, X, UV, visible, IR, microwave, radio.
- $E_0/B_0 = c$, not B_0/E_0.
- nT $= 10^{-9}$ T; MHz $= 10^{6}$ Hz.
- The energy densities of $\vec{E}$ and $\vec{B}$ are equal; don't double-count.

Question types: 1-mark: identify a wave from its use or λ, speed of EM waves, direction of $\vec{E}$ and $\vec{B}$. 2–3 marks: displacement current, wave equations. Case-based questions on the spectrum are common.

8F JEE extras

Useful results

- Average energy density: $u = \frac{1}{2}\varepsilon_0 E_0^2$ (half electric, half magnetic). Intensity $I = uc = \frac{1}{2}\varepsilon_0 E_0^2 c$.
- Radiation pressure: $\dfrac{I}{c}$ (absorbed), $\dfrac{2I}{c}$ (reflected); momentum $p = \dfrac{U}{c}$.
- Refractive index $n = \sqrt{\mu_r \varepsilon_r}$.

Example (Q8.10): $\lambda = \dfrac{3 \times 10^8}{2 \times 10^{10}} = 1.5$ cm; $B_0 = \dfrac{48}{3 \times 10^8} = 1.6 \times 10^{-7}$ T.

8G Pattern spotter

If you see...	Do this	See
Displacement current		
Charging capacitor	$i_d = \varepsilon_0\, d\Phi_E/dt = i_c$	Ex A
Rate of change of V	$dV/dt = I/C$	Ex A
Wave relations		
Wavelength from frequency	$\lambda = c/\nu$	Ex B
B_0 from E_0	$B_0 = E_0/c$	Ex B
Write $\vec{E}$, $\vec{B}$ equations	$\omega = 2\pi\nu$, $k = \omega/c$, $\vec{E} \perp \vec{B} \perp$ travel	Ex B
Photon energy	$E = h\nu$; $1240/\lambda$(nm) eV	Ex C
Identify a part of the spectrum	Use the λ table	8D